Ayush Kumar Pandey

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Summary

B.Tech ISE student (6th Sem, CGPA: 8.55/10.0), Presidency University, Bangalore. 2+ years experience in Java full-stack, machine learning, computer vision, and Android development. Proficient in AI-assisted development, prompt engineering, and building production systems using AI agents – consistently delivering faster, efficient, and token-optimised workflows. Keen interest in deepening expertise in LLMs and Generative AI. Seeking Software Engineering internship (7th Semester).

Education

Presidency University – B.Tech Information Science & Engineering — CGPA: 8.55/10.0 Bangalore
Relevant: DSA, OOP, AI & ML, Software Engineering, Cloud Computing, DBMS, Computer Networks *Sept 2023 – June 2027*

Technical Skills

Languages: Java, Python, JavaScript, SQL, HTML/CSS
Java & Frameworks: Spring Boot, JavaFX, Design Patterns (Factory/Strategy/Observer), Android Studio, Flask, React, RESTful APIs
ML & AI: TensorFlow, PyTorch, OpenCV, YOLOv8, ArcFace, ONNX Runtime, FAISS, CNN, Transfer Learning
AI Agents & Prompt Engineering: Agentic Workflow Design, System Architecture with AI, Prompt Optimisation, Token-Efficient Pipelines, n8n, UiPath RPA
Cloud, DB & Tools: AWS (EC2/S3), Docker, Git/GitHub, MySQL, SQLite, Selenium, Pandas, JIRA, Agile/Scrum

Experience

AI-Assisted Full-Stack Developer (Project-Based) 2023 – Present
Independent — Java, Python, Android Studio, Flask, React, AI Agents, Prompt Engineering
- Architected and shipped 12+ production projects by effectively leveraging AI agents and prompt engineering – designing optimised workflows and system architectures that maximise output while minimising token consumption.
- Built full-stack Java desktop and Android apps using MVC, OOP, and design patterns; developed 15+ math tools with custom 2D/3D graphing engine achieving 40% faster feature cycles.

Machine Learning & Computer Vision Developer (Project-Based) 2023 – Present
Independent — TensorFlow, PyTorch, OpenCV, ArcFace, FAISS
- Built end-to-end ML pipelines for facial recognition, ASL classification, and license plate detection at 88–94% accuracy; sole owner of FaceSort from ML architecture to packaged Windows installer.

Automation Developer (Project-Based) 2025
Independent — Python, Selenium, n8n, AI Workflow Automation
- Automated ERP scraping with 6-XPath fallback logic and built a faculty evaluation bot; designed AI-assisted automation workflows reducing task completion time from minutes to seconds.

Projects

FaceSort Desktop | *PySide6, ArcFace, ONNX, FAISS, SQLite, PyInstaller, AI-assisted architecture* Feb–Mar 2026
- Offline Windows desktop app with ArcFace 512-d embeddings and FAISS cosine similarity; confidence-based auto-assign/review/new-profile logic; MVC + multi-threading; NSIS single-click installer. System architecture designed with AI agent collaboration.

FaceSort v1.0 – Full-Stack Web App | *Flask, React, TypeScript, MTCNN, DeepFace, HDBSCAN, FAISS* Jan–Feb 2026
- RESTful Flask API + React SPA for facial recognition-based photo organisation; 1000+ images at 90% accuracy; 85% reduction in manual sorting via end-to-end ML pipeline.

Advanced Scientific Calculator | *Java, JavaFX, Factory/Strategy/Observer Patterns* 2024
- 15+ math tools with real-time 2D/3D graphing engine; modular OOP design delivers 40% faster development cycles.

Hand Sign Recognition System | *Python, TensorFlow, CNN, OpenCV* 2024
- 94% real-time ASL alphabet accuracy using custom CNN with transfer learning and live webcam inference.

Vehicle License Plate Logger & IoT Monitor | *Raspberry Pi, OpenCV, Tesseract OCR, Arduino* 2023
- 88% plate detection, sub-2s latency, 95% reduction in manual logging; 24/7 IoT server room sensing with cloud alerts.

Certifications

NPTEL – Introduction to Large Language Models (2026) | NOC26CS88S75020090604410485
NPTEL – Fundamentals of Virtual and Augmented Reality Systems (2026) | NOC26CS03S125020181504410485

Achievements

Shipped 12+ production projects independently by mastering AI-agent collaboration, prompt engineering, and token-efficient workflow design.
Sole developer for FaceSort Desktop: full product lifecycle from ML architecture design (with AI assistance) to packaged Windows installer.
Maintained 8.55 CGPA while delivering 88–94% ML accuracy and 80–95% effort reduction across all independent projects.